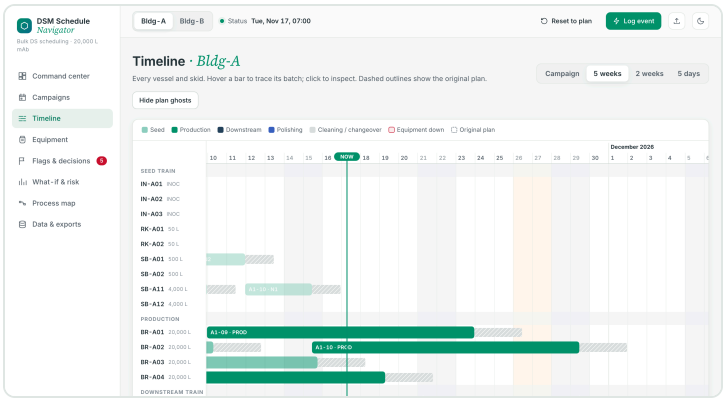
DSM Schedule *Navigator*

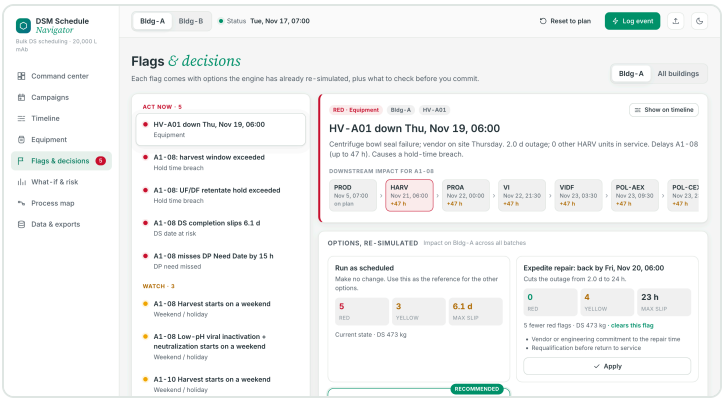
Scheduling and decision support for 20,000 L bulk drug substance manufacturing

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Live demo: jseggs.github.io/dsm-navigator

Bulk DS schedules are often kept in Excel and maintained by hand. This is a working tool that keeps Excel as the input but adds what a spreadsheet can't do well: when a window is missed, it shows where the delay lands downstream, which hold times are at risk, and which recovery options actually work.



Timeline: a harvest-skid outage ripples through the downstream train. Dashed outlines show the original plan.



Decision card: each option is re-simulated across every batch before the AD commits.

HOW IT WORKS

- **Excel is the backend.** 13 input sheets: settings, batch list, process template, polishing options, equipment, mass balance, campaigns, hold limits, flag rules, crew, materials, calendar, and a live event log. Every default carries a source / confidence note.
- **Finite-capacity scheduler.** Seed train, 4 x 20 kL bioreactors and one shared downstream train per building, with cleaning turnaround, transfer windows, hold limits, and weekday-only steps.
- **Durations from process data.** Protein A time is computed from titer, column size and DBC (5 cycles at 4 g/L, 7 at 5.5 g/L), so a titer change re-plans capture automatically.
- **Campaign planning to the DP Need Date.** Enter lots and the date Drug Product needs released DS; the tool back-schedules into open capacity and hands the batch list back to Excel.
- **Event logging and equipment board.** Delay, equipment down, batch lost, or hold extension; mark any unit down or out of service and the train re-plans around it. Steps already running are locked.
- **Red and yellow flags** for hold breaches, DS slips, DP Need Dates at risk, station load, weekend starts, crew shortfalls, materials and resin life.
- **Monte Carlo** on min / likely / max durations for P50 / P90 DS dates and breach probability.
- **Runs offline** as one HTML file. No install, no server, no data leaves the laptop.

WHAT IT SURFACED WITH ILLUSTRATIVE DATA

Harvest collision. A 48 h centrifuge outage pushes the next Bldg-A harvest past its 36 h window. The tool computes the latest repair time that avoids the breach (Fri 13:00), which cuts red flags from 5 to 0.

Holiday cascade. The same delay moves UF/DF onto Thanksgiving. With a weekday-only fill, the retentate waits 4.1 days against a 72 h hold, a second breach that is easy to miss by hand.

Cadence choice. Running Bldg-B at 4.5 instead of 4.25 days between thaws drops the simulated chance of any hold breach from 13% to about 0% and costs about 6 h at campaign end.

Resin life. Bldg-B Protein A resin runs out at batch B1-08. The tool finds the largest open window on the capture skid and schedules the re-pack there with no DS slip.

Independent concept by John Seggman. Built on public, industry-typical assumptions. Contains no proprietary data.